

Anomaly Inflow II

- Review Anomaly inflow Mechanism.
- 't Hooft anomalies $\rightarrow$ RG flow & anomaly matching conditions.
- Example: particle on a circle

see [arXiv:1905.09315](https://arxiv.org/abs/1905.09315)

action	Symmetry Group
$S = \int d^d x \mathcal{L}$	$G \times H$

Global symmetry $\rightarrow$ Gauge symmetry

Manifold

$X \rightarrow$ Principle bundle $G \rightarrow$ gauge group

insert background gauge field A

Ex. $\mathcal{L} = \partial_\mu \phi^* \partial^\mu \phi \quad U(1) \quad \phi \mapsto e^{i\alpha} \phi$

$$\mathcal{L} = (\partial_\mu + i g A_\mu) \phi^* (\partial_\mu - i g A_\mu) \phi$$

Anomalies

$G \times H$
 $\downarrow$ gauge symmetry
 $\downarrow$ global symmetry.

(original)
 symmetry types (G)

	Gauge	Global
Gauge transformation G.	X*	+ Noof
(Mixed) Global transformation H	e.g. ABJ break H to a (discrete) subgroup.	+ Noof e.g. Particle on a circle.

* 1. Anomaly inflow

2. Anomaly matching conditions $\rightarrow$ e.g. SM.

Anomaly Ansatz $\rightarrow$ e.g. Chiral Anomalies.

$$\underset{\text{partition fnc}}{Z[A]} \xrightarrow{G, H} Z[A'] = Z[A] \exp(-2\pi i \int_X \alpha(1, A))$$

Anomaly Cohomology

e.g. Regularisation Schemes $\rightarrow$ add counter terms to the action.

$\Rightarrow$ Partition fnc is subject to a regularisation scheme ambiguity.

$$Z[A] \sim Z[A] \exp(2\pi i \int_X \beta(A))$$

$\beta(A) \rightarrow$ counterterms.

Let the variation of β under $A \mapsto A^\lambda$ be: $\beta(A^\lambda) = \beta(A) - \delta_\lambda \beta(A)$

Then

$$\begin{aligned} Z[A^\lambda] &= \overset{\text{RHS}}{Z[A]} \exp(-2\pi i \int_X \alpha(\lambda, A)) \exp(-2\pi i \int_X \delta\beta(A)) \\ &\sim \underset{\text{LHS}}{Z[A]} \exp(-2\pi i \int_X \alpha(\lambda, A)) \end{aligned}$$

$$\Rightarrow \alpha \sim \alpha + \delta\beta$$

$$[\alpha] \in \{\alpha\} / (\alpha \sim \alpha + \delta\beta) \quad \text{i.e. defines an anomaly cohomology.}$$

Cohomology is discrete.

$\Rightarrow$ Anomaly equiv class is invariant under its deformations of the theory S .

- Ex
- smoothly varying Lagrangian parameters
 - Adding (infinitely) massive field.
 - RG flow

$\hookrightarrow [\alpha]$ invariant RG $\Rightarrow$ Anomaly matching
 $[\alpha]_{\text{IR}} = [\alpha]_{\text{UV}}$

Anomaly Inflow Mechanism

Purposes

1. counter balance gauge anomalies.
2. Measure $\forall$ Hooft anomalies.

$$S = \int_X \mathcal{L} \quad 2Y = X \quad A \rightarrow \tilde{A} \quad \tilde{A}|_X = A.$$



New bulk Lagrangian $\omega(\tilde{A})$

invertible field theory.

$$A[\tilde{A}] = \exp(2\pi i \int_X \omega(\tilde{A}))$$

1. want to
construct.

2. transforms as

$$A[\tilde{A}^\lambda] = A[\tilde{A}] \exp(2\pi i \int_X \alpha(\lambda, A))$$

$\Rightarrow$ Modified Partition fnc

Anomaly free.

$$\tilde{\mathcal{Z}}[A] = \mathcal{Z}[A] A[\tilde{A}] \Rightarrow \boxed{\tilde{\mathcal{Z}}[A^\lambda] = \mathcal{Z}[A]}$$

Anomaly Matching

implication of anomaly cohomology.

smooth deformations $\omega \xrightarrow{\mu} \omega_\mu$

$[\omega_\mu] \rightarrow$ equivalence classes.

invariant under continuous variations

- RG flow
- etc.

$Z[A] \xrightarrow{\mu} Z_\mu[A]$ characterised by $[\omega_\mu]$

Claim if $[\omega_\mu] \neq 0 \Rightarrow Z_\mu[A]$ along all of μ
will not invertible.

$\rightarrow$ Non-trivial theory.

Ex. RG.

UV $\xrightarrow{\mu=0}$ IR $\mu=\infty$

Typically Expect IR to be a
trivial theory No excitations
eg. true vacuum.

$[\omega_\mu] \neq 0 \Rightarrow$ • Gapless: massless interacting CFT
• Gapped: nontrivial TQFT.

$G \times H \rightarrow$ Global symmetry

$\Rightarrow$ gauged $G \Rightarrow [\omega_\mu] \neq 0 \Rightarrow$ Know that the spectrum
of the theory is non-trivial along
RG flow.

$$\omega_{\mu_1} \rightarrow [\omega_{\mu_1}]$$

$$\omega_{\mu_2} \rightarrow [\omega_{\mu_2}]$$

$\Rightarrow$ No way to deform one theory into the other.

Exception

$G \times H$ is spontaneously broken $\Rightarrow$ contain Goldstone modes
 $\Rightarrow$ nontrivial gapless.

key assumption to above: $G \times H$ is preserved along flow μ .

Example Particle on a Circle



$$\begin{aligned} x(T) &= x(0) \bmod 2\pi \\ &= x(0) + 2\pi n \quad n \in \mathbb{Z}. \end{aligned}$$

Classical Lagrangian.

$$L = \underbrace{\frac{1}{2} \dot{x}^2}_{\text{kinetic}} + \underbrace{\frac{\Theta}{2\pi} \dot{x}}_{\text{Magnetic field.}}$$

Two symmetries

$$U_1: x \mapsto x - \lambda \quad U(1)$$

$$C: x \mapsto -x \quad \mathbb{Z}_2$$

want to study the $\vee$ Hooft of $U_1 \Rightarrow$ gauge sym.

$$L_0 = \frac{1}{2} (\dot{x} + A_0)^2 + \frac{\Theta}{2\pi} (\dot{x} + A_0) \quad A_0 \mapsto A_0 + \frac{d\lambda}{dt}$$

Quantum Theory

$$Z[A] = \int D\alpha e^{i \int dt L_0}$$

Wick Rotate $t \mapsto \tau$ & add counterterm.

$$Z[A] = \int D\alpha e^{-\int d\tau L}$$

$$L = \frac{1}{2} (\dot{x} + A)^2 - i \frac{\Theta}{2\pi} (\dot{x} + A) \quad \underbrace{-ikA}_{\text{counter term.}}$$

compactify domain

$$\int d\tau L = \oint d\tau L \quad [0, T] \quad x(T) = x(0) + 2\pi$$

consequences

$$1. \quad \theta \sim \theta + 2\pi \quad [\theta \text{ const.}]$$

$$2. \quad k \in \mathbb{Z}.$$

Proof

$$\frac{1}{1} e^{\frac{i}{2\pi} \oint (\theta + \phi) \dot{x}} = e^{\frac{i}{2\pi} (\theta + \phi) (\underbrace{x(T) - x(0)}_{2\pi})} = e^{-i(\theta + \phi)} \Rightarrow \phi = 2\pi \quad \text{s.t. theory is invariant.}$$

Consider [invariant] shift of $\theta \mapsto \theta + \phi$

Proof:

$$e^{ik\oint A} \xrightarrow[\text{gauge transformation}]{A \mapsto A + \lambda} e^{ik\oint A + i k \oint \lambda} = e^{ik\oint A} e^{i k \oint \lambda} = e^{ik\oint A} e^{i k 2\pi} \Rightarrow k \in \mathbb{Z}.$$

For gauge invariance to hold.

$$e^{ik\oint A} \xrightarrow[\text{gauge transformation}]{A \rightarrow A + \lambda} e^{ik\oint A + i k \oint \underbrace{\lambda(t) - \lambda(0)}_{2\pi}} = e^{ik\oint A} e^{ik2\pi} \Rightarrow k \in \mathbb{Z}$$

For gauge invariance to hold.

L is gauge invariant. • $\dot{x} + A \mapsto \dot{x} - \dot{\lambda} + A + \dot{\lambda} = \dot{x} + A$ ✓

- $kA \sim kA + k\lambda$ since $k \in \mathbb{Z}$ ✓

What happens under C?

mixed 't Hooft anomaly

$$C: x \mapsto -x \quad A \mapsto -A$$

↓

Symmetry if $\theta = 0$ or $\theta = \pi$

$$\begin{aligned} L &\xrightarrow{C} L^C = \frac{1}{2}(\dot{x} + A)^2 + i \frac{\Theta}{2\pi} (\dot{x} + A) + ikA \\ &= \frac{1}{2}(\dot{x} + A)^2 - i \frac{\Theta}{2\pi} (\dot{x} + A) - ikA + i \frac{\Theta}{\pi} \dot{x} + i \frac{\Theta}{\pi} A + 2ikA \\ &= L + 2i \frac{\Theta}{2\pi} \dot{x} + i \left(\frac{2\Theta}{2\pi} + 2k \right) A \end{aligned}$$

$$e^{-\oint d\tau L} \mapsto e^{-\oint L} e^{-2i\frac{\theta}{2\pi}\oint d\tau i} \underbrace{e^{-i(2k + \frac{2\theta}{2\pi})\oint d\tau A}}_{e^{-2\pi i \int_X \kappa(\lambda, A)}} \quad \downarrow \quad c$$

$$\oint d\tau \dot{x} = x(T) - x(0) = 2\pi$$

$$e^{-2i\frac{\theta}{2\pi}\oint d\tau i} = e^{-2i\theta} = 1 \quad \text{if} \quad \underline{\theta = 0, \pi.}$$

Symmetry.

But anomalous phase

$$e^{-i(2k + \frac{2\theta}{2\pi})\oint d\tau A}$$

if $\theta = 0, \pi$ Then

$$Z[A] \mapsto Z[A^c] = Z[A] \exp(-i(2k + \frac{2\theta}{2\pi})\oint d\tau A)$$

- $\theta = 0 \Rightarrow (2k + \frac{2\theta}{2\pi}) = 2k \Rightarrow$ invariant if $k=0$
- $\theta = \pi \Rightarrow 2k + \frac{2\theta}{2\pi} = 2k+1 \Rightarrow$ invariant if $k = -\frac{1}{2}$.

$\hookrightarrow$ gauge invariance requires $k \in \mathbb{Z}$!

Anomaly Inflow

$$S' = X = \partial Y$$

$$F = d(\tilde{A} d\tau)$$

$$A[A] = \exp\left(\frac{i}{2} \int_Y F\right)$$

$$A \xrightarrow{\text{extend}} \tilde{A} \text{ s.t. } \tilde{A}|_X = A.$$

$X = 1\text{dim Manifold}$

$Y = 2\text{dim Manifold}$

needed s.t. $\exists$ curvature $F = dA$!

Note: on X , $d_X A = 0$

$$\exp\left(- (2k+1)i \int_{S^1} (\tilde{A} d\tau)\right) = \exp\left(- (2k+1)i \int_Y F\right)$$

$\underbrace{\int_{\partial Y} (\tilde{A} d\tau)}_{\int_Y d(\tilde{A} d\tau) = \int_Y F}$

$$\hat{Z}[A] = Z[A] \exp\left(2\pi i \int_Y \omega(\tilde{A})\right)$$

$$\omega(A) = K F \quad F \xrightarrow{C} -F$$

Under the (anomalous) C transformation

$$\hat{Z}[\tilde{A}^C] = Z[A^C] \exp\left(2\pi i \int_Y K(-F)\right)$$

$$= Z[A] \exp\left(-i(2k+1) \int_Y F\right) \exp\left(-2\pi i K \int_Y F\right)$$

$$= Z[A] \exp(2\pi i \int_Y K F)$$

$$\Rightarrow 2k+1 + K = -K$$

$$K = -\frac{1}{2}(2k+1)$$

$$\Rightarrow \tilde{Z}[\tilde{A}] = Z[A] \exp\left(-\frac{i}{2}(2k+1) \int_Y F\right)$$

Thus

$$\tilde{Z}[\tilde{A}] \text{ is } C\text{-invariant!} \quad \tilde{Z}[\tilde{A}^c] = \tilde{Z}[\tilde{A}]$$

Note: $Z[A] \rightarrow 1\text{dim anomalous theory}$

$\tilde{Z}[\tilde{A}] \rightarrow 2\text{dim Anomaly free theory}$

TODOs

- Compute deformation class $[u_\mu]$ of "particle on a circle".

- understand anomaly inflow picture for discrete symmetry

eg. $\hookrightarrow$ turned on by adding $\cos(2x)$ to Lagrangian $\Rightarrow$ expect to be in the same deformation class as above.

shift symmetry broken to
discrete gauge group
 $\mathbb{Z}_2 \times \mathbb{Z}_2$
 $\uparrow$
 C

- RG flow of "particle on a circle".

- What if? Gauge C & consider shift symmetries? [repeat above analysis]